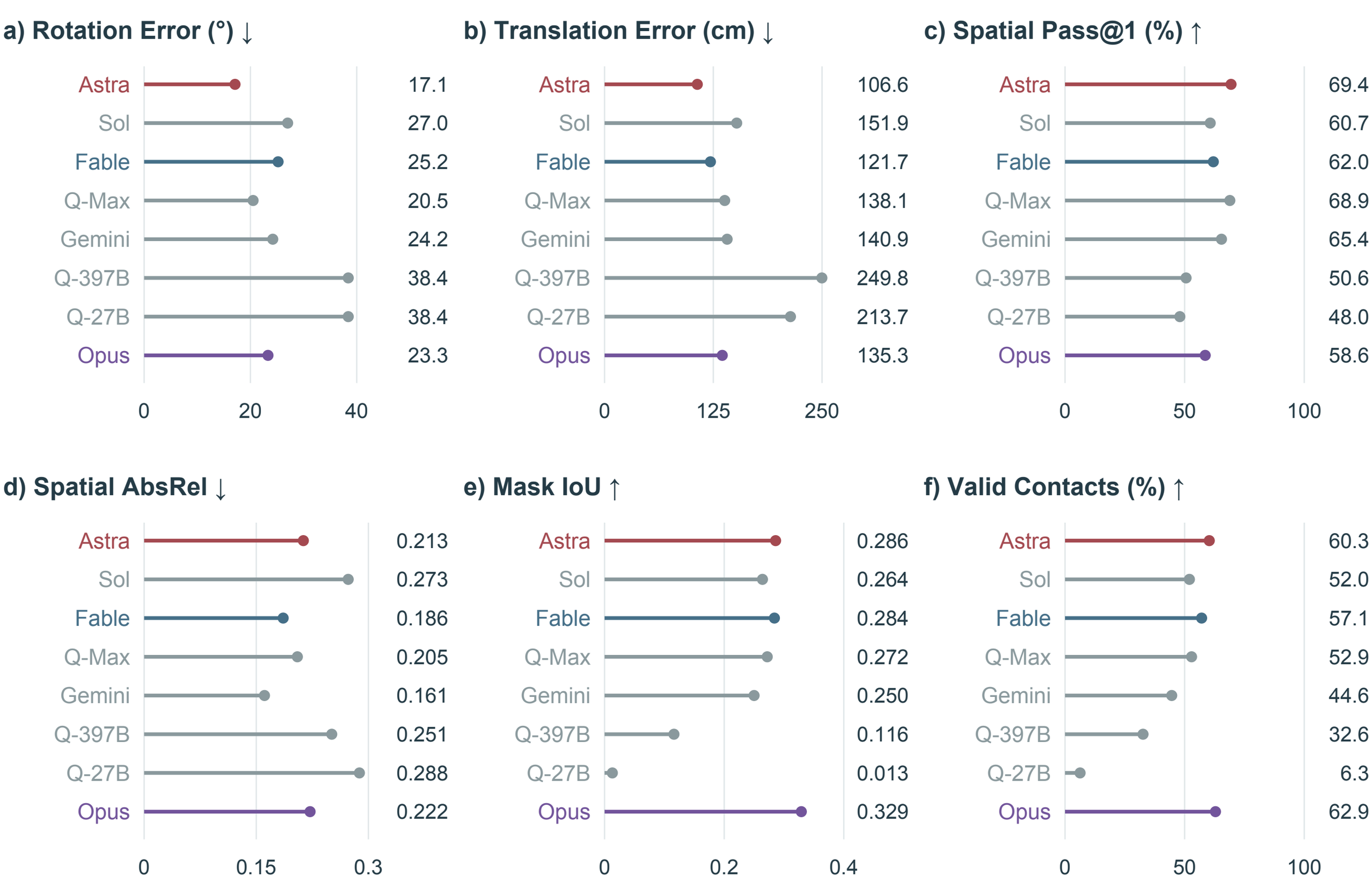

a) Rotation Error (°) ↓
Astra
Sol
Fable
Q-Max
Gemini
Q-397B
Q-27B
Opus
17.1
27.0
25.2
20.5
24.2
38.4
38.4
23.3
0
20
40
b) Translation Error (cm) ↓
Astra
Sol
Fable
Q-Max
Gemini
Q-397B
Q-27B
Opus
106.6
151.9
121.7
138.1
140.9
249.8
213.7
135.3
0
125
250
c) Spatial Pass@1 (%) ↑
Astra
Sol
Fable
Q-Max
Gemini
Q-397B
Q-27B
Opus
69.4
60.7
62.0
68.9
65.4
50.6
48.0
58.6
0
50
100
d) Spatial AbsRel ↓
Astra
Sol
Fable
Q-Max
Gemini
Q-397B
Q-27B
Opus
0.213
0.273
0.186
0.205
0.161
0.251
0.288
0.222
0
0.15
0.3
e) Mask IoU ↑
Astra
Sol
Fable
Q-Max
Gemini
Q-397B
Q-27B
Opus
0.286
0.264
0.284
0.272
0.250
0.116
0.013
0.329
0
0.2
0.4
f) Valid Contacts (%) ↑
Astra
Sol
Fable
Q-Max
Gemini
Q-397B
Q-27B
Opus
60.3
52.0
57.1
52.9
44.6
32.6
6.3
62.9
0
50
100